2026

Xavier's School for Gifted Shoppers

RETAIL CUSTOMER SHOPPING BEHAVIOR ANALYSIS

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SECTIONS TO REVIEW

01 Data Preparation Process

02 Sales by Age Groups

03 High-Value Discount Users

04 Top-Rated Products

05 Shipping Preferences and Their Impact

06 Subscription Impact

07 Customer Segmentation

DATASET OVERVIEW

3.9K Total Purchases

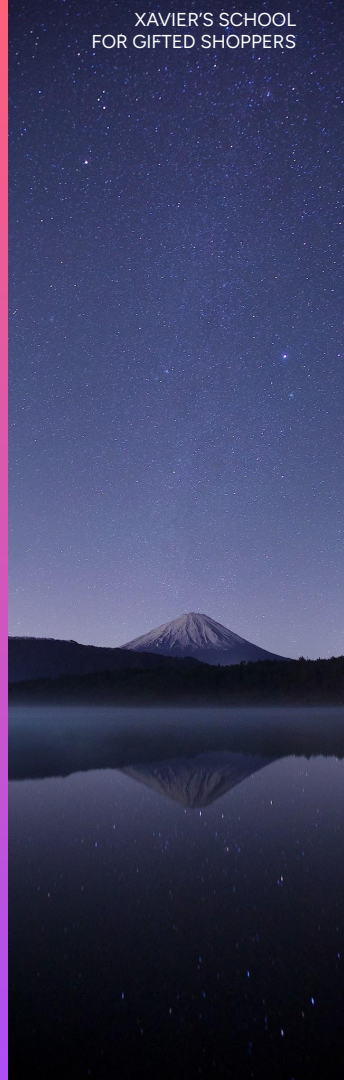
Comprehensive transaction data

18 Features

Customer demos & behavior

37 Missing Values

Only in 'Review Rating' Feature



Data Preparation Process

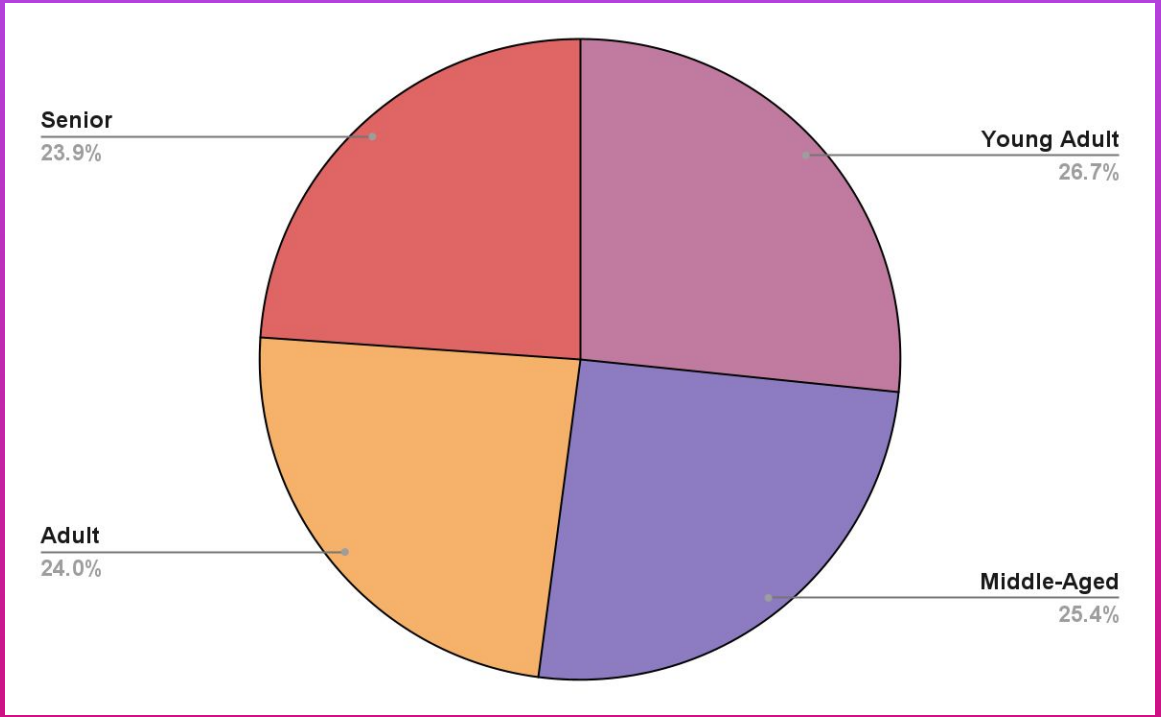
01	Data Loading	Imported dataset using pandas
02	Initial Exploration	Structure check and summary statistics
03	Missing Data Handling	Imputed "Review Rating" with median values for each respective category
04	Feature Engineering	Created columns for age groups and purchase frequency
05	Database Integration	Connected dataset to PostgreSQL for analysis

Key Insight

Young adult customers generate slightly higher total revenue than other age groups

Young adult-based marketing strategies could optimize revenue streams

Revenue by Age Groups



High-Value Discount Users

Customers using discounts while still spending above average

- Smart Shoppers
  - High spenders who maximize value with discounts
- Strategic Opportunity
  - Target premium customers with exclusive offers

Top-Rated Products

[1] GLOVES

3.86 out of 10

[2] SANDALS

3.84 out of 10

[3] BOOTS

3.82 out of 10

[4] HATS

3.80 out of 10

Shipping Preferences & Their Impact

EXPRESS SHIPPING

$61

Average Purchase Amount

STANDARD SHIPPING

$57

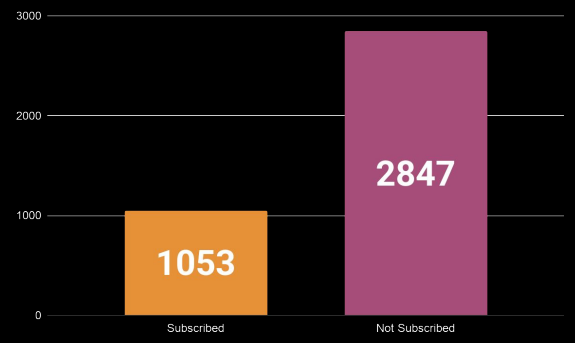
Average Purchase Amount

Express shipping customers spend, on average, 7% more per transaction than standard shipping customers

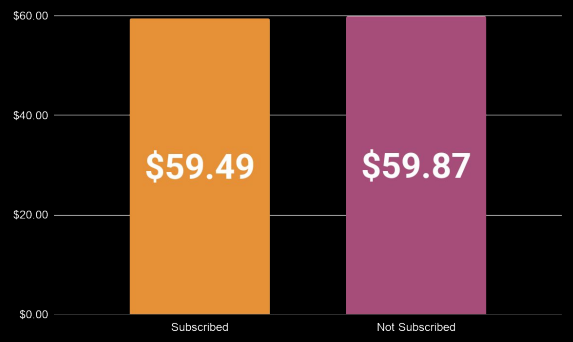
Find ways to incentivise customers to choose express shipping

Subscription Impact

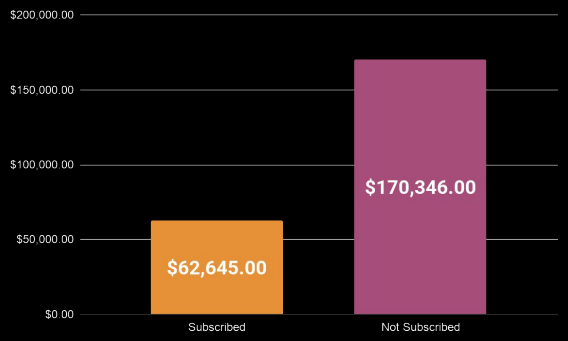
Subscribed vs. Not Subscribed Customers



Average Spend



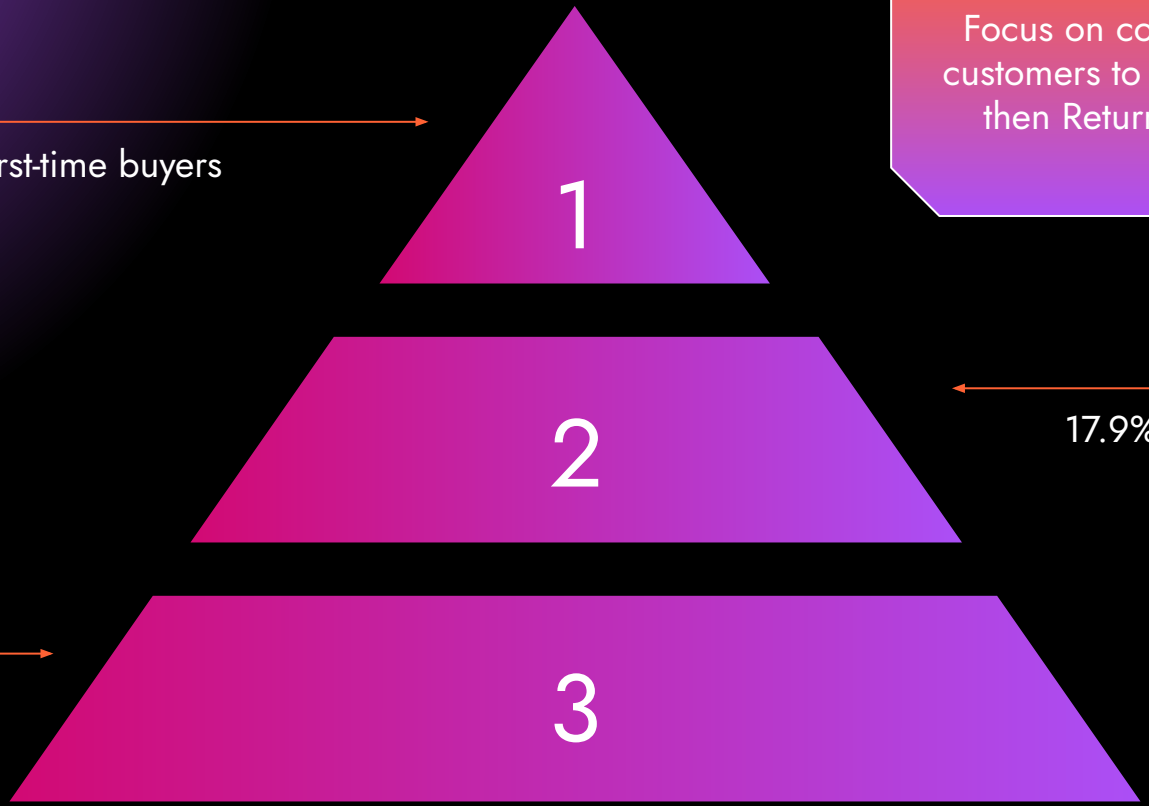
Total Revenue



Customer Segmentation

NEW

2.2% - First-time buyers



Focus on converting New customers to Returning, and then Returning to Loyal

RETURNING

17.9% - Regular shoppers

LOYAL

79.9% - High-value customers

Strategic Recommendations

Boost Subscriptions	Promote exclusive benefits for subscribers
Improve Loyalty Programs	Reward repeat buyers to increase retention
Targeted Marketing	Focus on high-revenue age groups and express users
Product Positioning	Highlight top-rated products in campaigns